Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/03/2006
PI Name:	Wang, Ning Ph.D.		
Project Title:	Control of Cell Mechanics and Function by Cell Microenvironment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	02115	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	08/01/2001	**End Date:**	10/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Butler, James (Harvard University)		
Grant/Contract No.:	NAG 2-1509		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The tasks of this project are: 1) to identify specific transmembrane molecules that transduce mechanical stress into alterations in endothelin-1 gene expression; 2) to determine to what degree endothelin-1 gene expression depends on the stiffness of the extracellular matrix; 3) to determine to what degree cytoskeletal tension and stiffness depends on the stiffness of the extracellular matrix and external mechanical stimulation. To understand how mechanical forces excert influences on cellular functions such as gene expression and the effects of extracellular matrix stiffness in cytoskeletal tension generation This is part of the overall effort in understanding cellular and molecular mechanisms of force-induced changes in cell functions		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The objective of this project is to elucidate how cell microenvironment controls and regulates cell functions. Elucidating the molecular pathway by which mechanical forces regulate gene expression and determining control of gene expression, cytoskeletal tension, and stiffness by the extracellular matrix mechanics should help us understand the molecular basis of mechanotransduction and mechanoregulation in microgravity as well as in phyiological and pathological conditions.
Task Progress:	Task Progress: During the last three months of no-cost extension of our project, we have finished all the proposed aims. In addition, one thing we examined is the effect of extracellular matrix stiffness on the tractions of living smooth muscle cells. We found that on soft substrate, cells generally generate lower tractions, consistent with published data in fibroblasts. However, surprisingly, we found that when cells come into contact, the tractions are higher at cell-cell interaction, opposite from what we would predict. We have also finished testing the 3D-magnetic twisting device. This new technology will allow us to probe details of how cells respond to directional forces never possible before. Peer Reviewed Original Reports: 1. Tolic-Norrelykke IM, Wang N. Traction in smooth muscle cells varies with cell spreading. J. Biomech. 38: 1405-1412, 2005. Invited Review Articles: 2. Stamenovic D, Wang N, Ingber, DE. Tensegrity models of cell-substrate interactions. In: Principles of Cellular Engineering: Understanding the Biomolecular Interface ed. M. R. King, Elsevier, Inc., 2006, pp. 81-101. 3. Stamenovic D, Wang N, Ingber DE. Tensegrity architecture and the mammalian Cell cytoskeleton. Ed by G. Sih. In press. Invited Symposia: January 2006 N. Wang, Invited Keynote Speaker at Annual Bioengineering Conference of Japanese Society of Mechanical Engineers
Bibliography Type:	Description: (Last Updated: 02/25/2021)
Articles in Peer-reviewed Journals	Tolic-Norrelykke IM, Wang N. "Traction in smooth muscle cells varies with cell spreading." J Biomech. 2005 Jul;38(7):1405-12. PMID: 15922751 , Jul-2005
Articles in Peer-reviewed Journals	An SS, Pennella CM, Gonnabathula A, Chen J, Wang N, Gaestel M, Hassoun PM, Fredberg JJ, Kayyali US. "Hypoxia alters biophysical properties of endothelial cells via p38 MAPK- and Rho kinase-dependent pathways." Am J Physiol Cell Physiol. 2005 Sep;289(3):C521-30. PMID: 15857906 , Sep-2005
Articles in Peer-reviewed Journals	Ingber DE, Wang N, Stamenovic D. "Tensegrity, cellular biophysics, and the mechanics of living systems." Rep Prog Phys. 2014 Apr;77(4):046603. https://doi.org/10.1088/0034-4885/77/4/046603 ; PMID: 24695087; PMCID: PMC4112545 , Apr-2014
Books/Book Chapters	Stamenovic D, Wang N, Ingber DE. "Tensegrity architecture and the mammalian cell cytoskeleton." in "Multiscaling in Molecular and Continuum Mechanics: Interaction of Time and Size from Macro to Nano." Ed. G.C. Sih. Dordrecht : Springer, 2007. p. 321-338., Jan-2007
Books/Book Chapters	Stamenovic D, Wang N, Ingber, DE. "Tensegrity models of cell-substrate interactions." in "Principles of Cellular Engineering: Understanding the Biomolecular Interface." Ed. M.R. King. Amsterdam : Elsevier, Inc., 2006., Jan-2006